Lecture 33 (*May 14*)

Reading: none
Viewing: none

Statistics is the sibling of probability, but they differ slightly in their goals. While probability anticipates the likelihood of future events, statistics interprets data from past events. Let's say we are performing an experiment that measures a certain quantity. We run the experiment n times and we get the following values, called **samples**:

$$x_1, \ldots, x_n \tag{305}$$

The collection of samples is often called the **data set**.

Definition 29. *The **mean** of the data set* (305) *is:*

$$\mu = \frac{1}{n}(x_1 + \cdots + x_n) \tag{306}$$

*The **variance** of the data set* (305) *is:*

$$\Sigma = \frac{1}{n-1}\Big[(x_1 - \mu)^2 + \cdots + (x_n - \mu)^2\Big] \tag{307}$$

*The **standard deviation** is the square root of the variance:* $\sigma = \sqrt{\Sigma}$.

As $n \to \infty$, the quantities (306) and (307) tend to the mean and variance of the underlying probability distribution of whatever we're measuring. For example, if you're measuring a quantity which naturally obeys a normal distribution (e.g. the numeric grades in a large class), then the graph of the function "$f(x)$ = number of students who got grade x" will look like a bell curve.

Remark. *You might wonder why we're using the denominator $n-1$ in* (307) *instead of n, which might seem much more appropriate from a probability point of view. The discrepancy, namely $\frac{n}{n-1}$ is called **Bessel's correction**, and we will explain it from the point of view of linear algebra later on. It is not so significant when n is large, but it makes a difference when n is small.*

Explicitly, the mean is just measuring the average of the samples in question, while the variance is measuring how far the samples are from their mean. These notions can be presented in the language of linear algebra, by introducing the vectors:

$$\boldsymbol{o} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

Then the mean is equal to:

$$\mu = \frac{\boldsymbol{o}^T \boldsymbol{x}}{\boldsymbol{o}^T \boldsymbol{o}} \tag{308}$$

and thus $\boldsymbol{o}\mu$ can be interpreted as the projection of the vector of samples $\boldsymbol{x}$ onto the vector $\boldsymbol{o}$. Meanwhile, the variance is:

$$\Sigma = \frac{||\boldsymbol{x} - \boldsymbol{o}\mu||^2}{n-1} = \frac{||P\boldsymbol{x}||^2}{n-1} \tag{309}$$

where $P = I - \frac{\boldsymbol{o}\boldsymbol{o}^T}{\boldsymbol{o}^T\boldsymbol{o}}$ is the projection matrix onto the orthogonal complement to $\boldsymbol{o}$. This orthogonal complement is an $n-1$ dimensional vector space, and indeed we have:

$$P = \begin{bmatrix} \frac{n-1}{n} & \cdots & -\frac{1}{n} \\ \vdots & \ddots & \vdots \\ -\frac{1}{n} & \cdots & \frac{n-1}{n} \end{bmatrix} \tag{310}$$

It is elementary to show that the diagonalization of P is:

$$P = QDQ^T, \quad \text{where } Q \text{ is orthogonal and } D = \begin{bmatrix} 1 & \dots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & 1 & 0 \\ 0 & \dots & 0 & 0 \end{bmatrix} \tag{311}$$

Therefore, if the n entries of the sample vector $\boldsymbol{x}$ are drawn from a normal probability distribution with mean 0 and variance 1, then so will the entries of the vector $\boldsymbol{y} = Q^T\boldsymbol{x}$ (this is because multiplying by orthogonal matrices does not change length). But we have:

$$||P\boldsymbol{x}||^2 = ||QDQ^T\boldsymbol{x}||^2 = ||D\boldsymbol{y}||^2 = y_1^2 + \cdots + y_{n-1}^2 \tag{312}$$

and so the variance of (312) is $n-1$. We have just shown that the variance of the quantity $||P\boldsymbol{x}||^2$ is $\frac{n-1}{n}$ times that of the quantity $||\boldsymbol{x}||^2$, and Bessel's correction is simply a way to correct for this.

Now assume we have two data sets: $x_1, \dots, x_n$ and $y_1, \dots, y_n$ (say age and height) and we want to see how correlated they are. As in the case of probability, the appropriate notion is the **covariance**:

$$\Sigma_{xy} = \frac{1}{n-1}\Big[(x_1-\mu)(y_1-\nu) + \cdots + (x_n-\mu)(y_n-\nu)\Big] \tag{313}$$

where μ and ν are the means of the samples $x_1, \dots, x_n$ and $y_1, \dots, y_n$, respectively. Note that Σ_{xx} is just the variance of the sample set $x_1, \dots, x_n$, so the reason why we use the denominator $n-1$ instead of n in (313) is still Bessel's correction, as explained above. In terms of vectors:

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

the covariance takes the form:

$$\boxed{\Sigma_{xy} = \frac{\boldsymbol{x}^T P \boldsymbol{y}}{n-1}} \tag{314}$$

since $P^2 = P$, $P^T = P$ (being a projection matrix) and $P\boldsymbol{x} = \begin{bmatrix} x_1-\mu \\ \vdots \\ x_n-\mu \end{bmatrix}$, $P\boldsymbol{y} = \begin{bmatrix} y_1-\nu \\ \vdots \\ y_n-\nu \end{bmatrix}$. Also:

$$|\Sigma_{xy}| \leq \sqrt{\Sigma_{xx}\Sigma_{yy}} \tag{315}$$

just like in the case of probability. The covariance is greatest (or lowest) when x and y are the most correlated (anti-correlated). If the two data sets are independent, then you would expect the

covariance to be very close to 0, and actually tend to 0 as you take more and more samples $n \to \infty$.

Finally, assume we have m data sets, so let's put them in an $n \times m$ matrix:

$$\boldsymbol{A} = \begin{bmatrix} x_1 & y_1 & z_1 & \dots \\ \vdots & \vdots & \vdots & \ddots \\ x_n & y_n & z_n & \dots \end{bmatrix}$$

If the x's, y's, z's ... have means μ, ν, ξ ..., then:

$$P\boldsymbol{A} = \begin{bmatrix} x_1 - \mu & y_1 - \nu & z_1 - \xi & \dots \\ \vdots & \vdots & \vdots & \ddots \\ x_n - \mu & y_n - \nu & z_n - \xi & \dots \end{bmatrix} \tag{316}$$

The **covariance matrix** of the data sets is the $m \times m$ matrix defined by:

$$K = \begin{bmatrix} \Sigma_{xx} & \Sigma_{xy} & \Sigma_{xz} & \dots \\ \Sigma_{yx} & \Sigma_{yy} & \Sigma_{yz} & \dots \\ \Sigma_{zx} & \Sigma_{zy} & \Sigma_{zz} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \tag{317}$$

and is given by the formula:

$$\boxed{K = \frac{\boldsymbol{A}^T P \boldsymbol{A}}{n-1}} \tag{318}$$

To see this, just note for instance that the $(1,2)$ entry of the matrix identity (318) is precisely (314).

Principal component analysis (PCA) involves diagonalizing the covariance matrix. If:

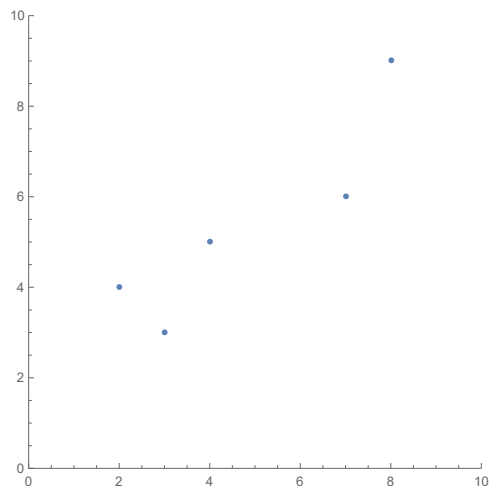
$$K = QDQ^T$$

with Q orthogonal and D diagonal (with positive diagonal entries) then (318) reads:

$$QDQ^T = \frac{\boldsymbol{A}^T P \boldsymbol{A}}{n-1} \quad \Rightarrow \quad D = \frac{(\boldsymbol{A}Q)^T P (\boldsymbol{A}Q)}{n-1}$$

This means that the data set $\boldsymbol{B} = \boldsymbol{A}Q$ (whose columns are linear combinations of the constituent data sets of $\boldsymbol{A}$) has diagonal covariance. Therefore, the constituent data sets of $\boldsymbol{B}$ are mutually uncorrelated. Moreover, if the i–th diagonal entry of D is the largest (respectively smallest) then the i–th constituent data set of $\boldsymbol{B}$ has the largest (respectively smallest) variance. Therefore, this approach allows us to isolate those linear combinations in a collection of data sets which produce the most variance from those which produce the least variance. For example, consider the plot: of two data sets: the entries of $\boldsymbol{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_5 \end{bmatrix}$ are on the horizontal axis and the entries of $\boldsymbol{y} = \begin{bmatrix} x_1 \\ \vdots \\ x_5 \end{bmatrix}$ are on the vertical axis. Explicitly, they are given by

$$\boldsymbol{x} = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 7 \\ 8 \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{y} = \begin{bmatrix} 4 \\ 3 \\ 5 \\ 6 \\ 9 \end{bmatrix}$$



and suppose we wish to compute their covariance matrix, and to apply PCA to them (i.e. to find linear combinations of the data sets which are mutually uncorrelated, and to compute the variance of those). Start by constructing the matrix:

$$\boldsymbol{A} = \left[\ \boldsymbol{x} \mid \boldsymbol{y}\ \right] = \begin{bmatrix} 2 & 4 \\ 3 & 3 \\ 4 & 5 \\ 7 & 6 \\ 8 & 9 \end{bmatrix}$$

Then compute:

$$P\boldsymbol{A} = \frac{1}{5}\begin{bmatrix} -14 & -7 \\ -9 & -12 \\ -4 & -2 \\ 11 & 3 \\ 16 & 18 \end{bmatrix} \tag{319}$$

Then the covariance matrix is given by formula (318):

$$K = \frac{1}{20}\begin{bmatrix} 134 & 107 \\ 107 & 106 \end{bmatrix}$$

You could perform PCA by diagonalizing the matrix K, or you could achieve the same result by instead computing the SVD of the matrix $P\boldsymbol{A}$. Let's opt for the latter:

$$P\boldsymbol{A} = U\Sigma V^T \qquad \text{where} \qquad \Sigma = \begin{bmatrix} \sqrt{24 + \frac{1}{5}\sqrt{11645}} & 0 \\ 0 & \sqrt{24 - \frac{1}{5}\sqrt{11645}} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

and U and V are orthogonal matrices. Then the covariance matrix is given by:

$$K = \frac{V\Sigma^T U^T U \Sigma V^T}{5-1} = V \begin{bmatrix} 6 + \frac{1}{20}\sqrt{11645} & 0 \\ 0 & 6 - \frac{1}{20}\sqrt{11645} \end{bmatrix} V^T \tag{320}$$

The meaning of this is the following: the entries of the 2×2 orthogonal matrix V teach you how to construct mutually independent combinations of the data sets $\boldsymbol{x}$ and $\boldsymbol{y}$. Explicitly, if:

$$V = \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix}$$

then the data sets:

$$\left[\ \boldsymbol{x}' \ \middle|\ \boldsymbol{y}' \ \right] = \left[\ \boldsymbol{x} \ \middle|\ \boldsymbol{y} \ \right] V \qquad \Leftrightarrow \qquad \begin{cases} \boldsymbol{x}' = v_{11}\boldsymbol{x} + v_{21}\boldsymbol{y} \\ \boldsymbol{y}' = v_{12}\boldsymbol{x} + v_{22}\boldsymbol{y} \end{cases}$$

are mutually uncorrelated. Moreover, the variances of $\boldsymbol{x}'$ and $\boldsymbol{y}'$ are the two numbers on the diagonal of (320), and numerically they are roughly equal to 11.39 and 0.6. This implies that the linear combination $\boldsymbol{y}'$ of our two data sets has very little variance, and so it is roughly constant. Meanwhile, the linear combination $\boldsymbol{x}'$ is very "noisy", since it has rather high variance.